

Chemical Bonding

1. A chemical bond is the force that

- A) Separates atoms
- B) Holds atoms together
- C) Breaks molecules
- D) Creates energy

Answer: B

2. Chemical bonding occurs to achieve

- A) Stability
- B) Instability
- C) High energy
- D) Pressure

Answer: A

3. Atoms combine to achieve

- A) Noble gas configuration
- B) Large size
- C) High density
- D) Colour

Answer: A

4. The outermost shell electrons are called

- A) Core electrons
- B) Valence electrons
- C) Protons
- D) Neutrons

Answer: B

5. Valency refers to

- A) Number of protons
- B) Combining capacity of atom
- C) Atomic mass
- D) Number of neutrons

Answer: B

6. An atom that loses electrons forms

- A) Anion
- B) Cation
- C) Molecule
- D) Radical

Answer: B

7. An atom that gains electrons forms

- A) Cation
- B) Anion
- C) Atom
- D) Molecule

Answer: B

8. Ionic bond is formed by

- A) Sharing electrons
- B) Transfer of electrons
- C) Breaking atoms
- D) Splitting nucleus

Answer: B

9. Covalent bond is formed by

- A) Transfer of electrons
- B) Sharing of electrons
- C) Losing protons
- D) Gaining neutrons

Answer: B

10. Ionic compounds are generally

- A) Soft
- B) Hard and brittle
- C) Liquid
- D) Gas

Answer: B

11. Ionic compounds have

- A) Low melting point

- B) High melting point
- C) Medium melting point
- D) No melting point

Answer: B

12. Ionic compounds conduct electricity in

- A) Solid state
- B) Molten or aqueous state
- C) Gas state
- D) Vacuum

Answer: B

13. Example of ionic compound

- A) H_2O
- B) CO_2
- C) NaCl
- D) CH_4

Answer: C

14. Sodium chloride is formed by

- A) Sharing electrons
- B) Transfer of electrons
- C) Losing neutrons
- D) Gaining protons

Answer: B

15. Covalent compounds usually have

- A) High melting point
- B) Low melting point
- C) Very high melting point
- D) Infinite melting point

Answer: B

16. Example of covalent compound

- A) NaCl
- B) MgO
- C) H_2O

D) CaCl_2

Answer: C

17. Hydrogen molecule forms

A) Ionic bond

B) Covalent bond

C) Metallic bond

D) Hydrogen bond

Answer: B

18. Oxygen molecule contains

A) Single bond

B) Double bond

C) Triple bond

D) Ionic bond

Answer: B

19. Nitrogen molecule contains

A) Single bond

B) Double bond

C) Triple bond

D) Ionic bond

Answer: C

20. Octet rule states atoms tend to have

A) 2 electrons

B) 4 electrons

C) 8 electrons in outer shell

D) 16 electrons

Answer: C

21. Helium is stable with

A) 2 electrons

B) 8 electrons

C) 10 electrons

D) 4 electrons

Answer: A

22. Noble gases are stable because they have

- A) Complete valence shell
- B) Incomplete shell
- C) High mass
- D) Large size

Answer: A

23. Metallic bond occurs between

- A) Metals
- B) Non-metals
- C) Metals and non-metals
- D) Gases

Answer: A

24. Metals conduct electricity due to

- A) Free electrons
- B) Protons
- C) Neutrons
- D) Molecules

Answer: A

25. A molecule is

- A) Single atom
- B) Group of atoms chemically bonded
- C) Proton
- D) Electron

Answer: B

26. The smallest particle of compound is

- A) Atom
- B) Molecule
- C) Proton
- D) Neutron

Answer: B

27. Hydrogen has valency

- A) 1

B) 2

C) 3

D) 4

Answer: A

28. Oxygen has valency

A) 1

B) 2

C) 3

D) 4

Answer: B

29. Nitrogen has valency

A) 1

B) 2

C) 3

D) 4

Answer: C

30. Carbon has valency

A) 1

B) 2

C) 3

D) 4

Answer: D

31. Sodium forms

A) Na^+

B) Na^-

C) Na^{2+}

D) Na^{2-}

Answer: A

32. Chlorine forms

A) Cl^+

B) Cl^-

C) Cl^{2+}

D) Cl^{2-}

Answer: B

33. Ionic bond forms between

A) Two metals

B) Two non-metals

C) Metal and non-metal

D) Noble gases

Answer: C

34. Covalent bond forms between

A) Two non-metals

B) Two metals

C) Metal and non-metal

D) Noble gases

Answer: A

35. Hydrogen molecule H_2 has

A) One covalent bond

B) Two bonds

C) Three bonds

D) Four bonds

Answer: A

36. Methane (CH_4) has

A) One bond

B) Two bonds

C) Three bonds

D) Four bonds

Answer: D

37. Water molecule has

A) One bond

B) Two bonds

C) Three bonds

D) Four bonds

Answer: B

38. Carbon dioxide contains

- A) Single bond
- B) Double bond
- C) Triple bond
- D) Ionic bond

Answer: B

39. The formula of ammonia is

- A) NH_3
- B) NH_2
- C) NH_4
- D) NH

Answer: A

40. The formula of methane is

- A) CH_3
- B) CH_4
- C) C_2H_6
- D) CO_2

Answer: B

41. The formula of water is

- A) H_2O
- B) H_2O_2
- C) HO
- D) H_3O

Answer: A

42. Covalent bonds involve

- A) Transfer of electrons
- B) Sharing electrons
- C) Transfer of protons
- D) Sharing neutrons

Answer: B

43. Ionic bonds involve

- A) Electron transfer

- B) Electron sharing
- C) Proton transfer
- D) Neutron sharing

Answer: A

44. Metals lose electrons because they have

- A) Low ionization energy
- B) High ionization energy
- C) High density
- D) High mass

Answer: A

45. Non-metals gain electrons because they have

- A) High electronegativity
- B) Low density
- C) High mass
- D) Low charge

Answer: A

46. Ionic compounds dissolve in

- A) Water
- B) Oil
- C) Petrol
- D) Gas

Answer: A

47. Covalent compounds dissolve in

- A) Water
- B) Organic solvents
- C) Salt
- D) Sugar

Answer: B

48. Ionic compounds form

- A) Molecules
- B) Crystal lattice
- C) Gas

D) Liquid only

Answer: B

49. Sodium chloride crystal contains

A) Molecules

B) Ions

C) Atoms only

D) Protons

Answer: B

50. Chemical bond forms due to

A) Attraction between atoms

B) Repulsion

C) Heat

D) Pressure

Answer: A

51. Bond energy refers to

A) Energy needed to break bond

B) Energy to form bond

C) Energy to heat atom

D) Energy to cool atom

Answer: A

52. Bond formation releases

A) Energy

B) Heat only

C) Light only

D) Pressure

Answer: A

53. Bond breaking requires

A) Energy

B) Cooling

C) Pressure

D) Vacuum

Answer: A

54. Noble gases rarely form bonds because they

- A) Are stable
- B) Are reactive
- C) Are metals
- D) Are liquids

Answer: A

55. Hydrogen bond is

- A) Weak attraction
- B) Strong ionic bond
- C) Metallic bond
- D) Nuclear bond

Answer: A

56. Hydrogen bonding occurs in

- A) Water
- B) Sodium chloride
- C) Methane
- D) Carbon dioxide

Answer: A

57. Metallic bonding gives metals

- A) Conductivity
- B) Insulation
- C) Brittleness
- D) Non-reactivity

Answer: A

58. Metallic bonds allow metals to be

- A) Malleable
- B) Brittle
- C) Gas
- D) Insulators

Answer: A

59. Ductility of metals is due to

- A) Metallic bonding

- B) Ionic bonding
- C) Covalent bonding
- D) Hydrogen bonding

Answer: A

60. Sharing one pair of electrons forms

- A) Single bond
- B) Double bond
- C) Triple bond
- D) Ionic bond

Answer: A

61. Sharing two pairs forms

- A) Single bond
- B) Double bond
- C) Triple bond
- D) Metallic bond

Answer: B

62. Sharing three pairs forms

- A) Single bond
- B) Double bond
- C) Triple bond
- D) Ionic bond

Answer: C

63. Molecules with covalent bonds are usually

- A) Poor conductors
- B) Good conductors
- C) Metals
- D) Ionic

Answer: A

64. Ionic bonds are

- A) Strong electrostatic attraction
- B) Weak attraction
- C) No attraction

D) Magnetic attraction

Answer: A

65. Sodium has valency

A) 1

B) 2

C) 3

D) 4

Answer: A

66. Magnesium has valency

A) 1

B) 2

C) 3

D) 4

Answer: B

67. Aluminium has valency

A) 1

B) 2

C) 3

D) 4

Answer: C

68. Carbon forms

A) Covalent bonds

B) Ionic bonds

C) Metallic bonds

D) No bonds

Answer: A

69. Hydrogen needs ____ electrons to complete duplet

A) 1

B) 2

C) 3

D) 4

Answer: A

70. Oxygen needs ____ electrons to complete octet

- A) 1
- B) 2
- C) 3
- D) 4

Answer: B

71. Nitrogen needs ____ electrons to complete octet

- A) 1
- B) 2
- C) 3
- D) 4

Answer: C

72. Carbon needs ____ electrons to complete octet

- A) 1
- B) 2
- C) 3
- D) 4

Answer: D

73. Chemical bond forms when atoms

- A) Gain stability
- B) Lose stability
- C) Lose energy only
- D) Gain mass

Answer: A

74. Ionic compounds are usually

- A) Solids
- B) Liquids
- C) Gases
- D) Plasma

Answer: A

75. Covalent compounds are usually

- A) Liquids or gases

- B) Metals
- C) Solids only
- D) Plasma

Answer: A

76. Chemical formula shows

- A) Elements present
- B) Ratio of atoms
- C) Both
- D) None

Answer: C

77. NaCl contains

- A) Na^+ and Cl^- ions
- B) Na atoms only
- C) Cl atoms only
- D) Molecules

Answer: A

78. H_2 molecule contains

- A) 2 hydrogen atoms
- B) 1 hydrogen atom
- C) 3 hydrogen atoms
- D) 4 hydrogen atoms

Answer: A

79. O_2 molecule contains

- A) 1 oxygen atom
- B) 2 oxygen atoms
- C) 3 oxygen atoms
- D) 4 oxygen atoms

Answer: B

80. N_2 molecule contains

- A) 1 nitrogen atom
- B) 2 nitrogen atoms
- C) 3 nitrogen atoms

D) 4 nitrogen atoms

Answer: B

81. Chemical bonding explains

A) Formation of compounds

B) Properties of matter

C) Structure of molecules

D) All of these

Answer: D

82. Bonds form because atoms

A) Lower energy

B) Increase energy

C) Increase mass

D) Decrease size

Answer: A

83. Ionic bonds form crystals due to

A) Electrostatic forces

B) Gravity

C) Magnetism

D) Pressure

Answer: A

84. Covalent molecules have

A) Fixed shapes

B) No shapes

C) Infinite size

D) No bonds

Answer: A

85. Metallic bonding involves

A) Sea of electrons

B) Protons

C) Neutrons

D) Molecules

Answer: A

86. Bond angle refers to

- A) Angle between bonds
- B) Length of bond
- C) Mass of bond
- D) Colour of bond

Answer: A

87. Bond length refers to

- A) Distance between nuclei
- B) Energy of bond
- C) Mass of bond
- D) Angle of bond

Answer: A

88. Ionic bond strength depends on

- A) Charge of ions
- B) Distance between ions
- C) Both
- D) None

Answer: C

89. Chemical bonding is studied in

- A) Chemistry
- B) Physics
- C) Biology
- D) Astronomy

Answer: A

90. Bonds determine

- A) Structure
- B) Properties
- C) Reactivity
- D) All of these

Answer: D

91. HCl molecule has

- A) Covalent bond

- B) Ionic bond
- C) Metallic bond
- D) Hydrogen bond

Answer: A

92. CO₂ molecule has

- A) Single bonds
- B) Double bonds
- C) Triple bonds
- D) Ionic bonds

Answer: B

93. NH₃ molecule has

- A) Three covalent bonds
- B) Two bonds
- C) One bond
- D) Four bonds

Answer: A

94. CH₄ molecule forms

- A) Four covalent bonds
- B) Three bonds
- C) Two bonds
- D) One bond

Answer: A

95. Chemical bonding involves

- A) Electrons
- B) Protons
- C) Neutrons
- D) Nucleus only

Answer: A

96. Valence electrons determine

- A) Chemical properties
- B) Colour
- C) Density

D) Mass

Answer: A

97. Ionic bond results in

A) Formation of ions

B) Molecules only

C) Atoms only

D) Neutrons

Answer: A

98. Covalent bond results in

A) Molecules

B) Ions

C) Protons

D) Neutrons

Answer: A

99. Stability of atom is achieved by

A) Completing outer shell

B) Removing nucleus

C) Adding neutrons

D) Removing protons

Answer: A

100. Chemical bonding helps atoms

A) Form compounds

B) Stay separate

C) Break apart

D) Disappear

Answer: A